

Lecture 9

- Electric (Scalar) Potential
- Examples

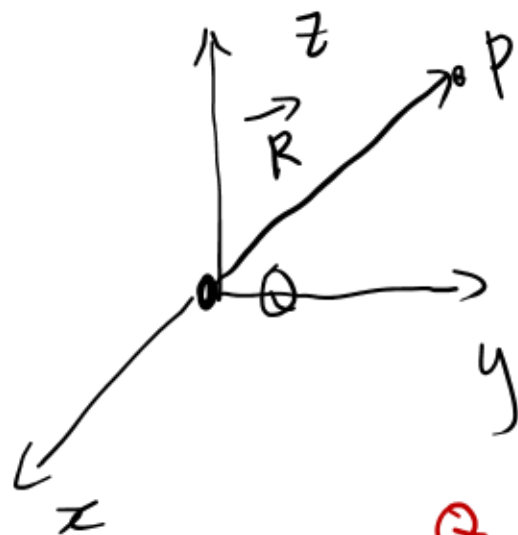
Midterm 1 information

Time: Monday October 5, in-class (75 mins)

Cover: everything from Lecture 01 to Lecture 11. [Midterm 1 Equation Sheets \(provided with your exam\)](#) ↓

- You can take one A4 (both sides) sheet of notes. Other notes/books are not permitted.
- You can use a non-graphing calculator. Use of other electronic devices is not permitted.
- Most credit is for setting up the problem and showing your work. Do not leave problems blank! If you get stuck or slowed down, move on.

Review

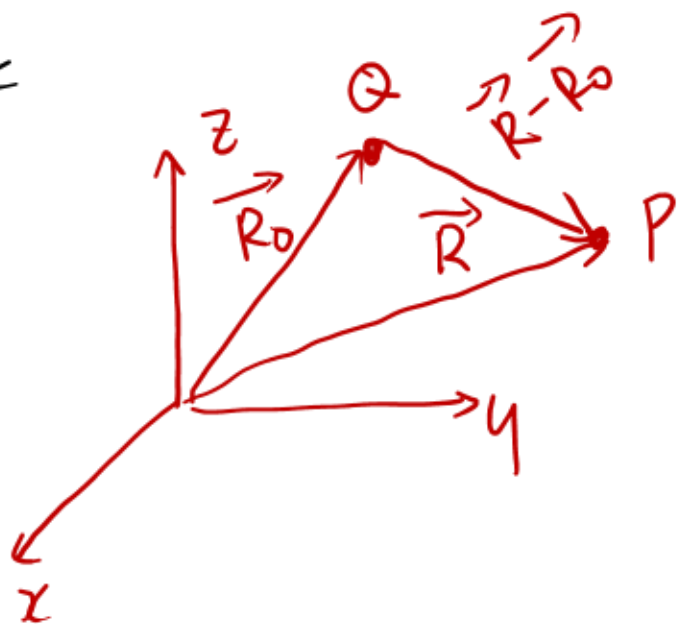


$$\vec{R} = R \hat{R}$$

with ref $\rightarrow \infty$

$$\underline{V(R)} = \frac{Q}{4\pi\epsilon(R)}$$

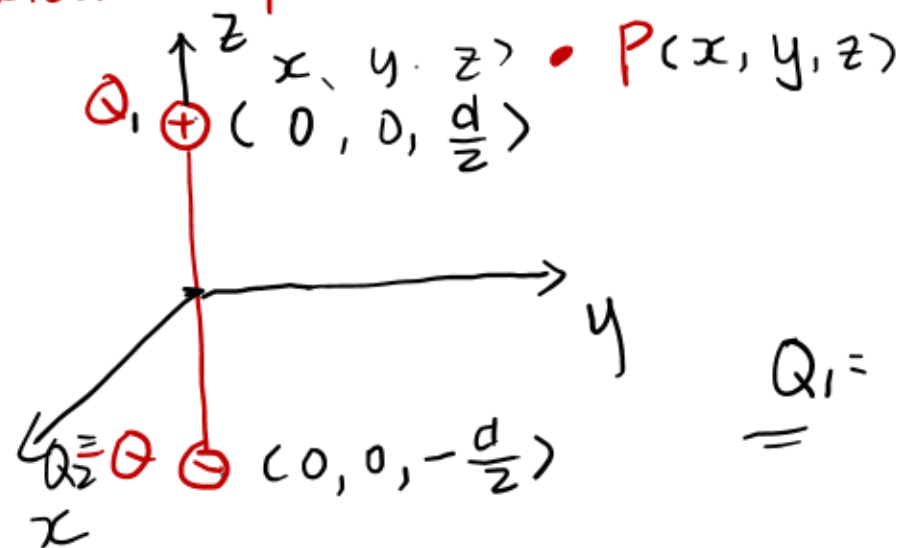
distance from point of obs. to source Q



$$V_P = \frac{Q}{4\pi\epsilon |\vec{R} - \vec{R}_0|}$$

Electric potential due to multiple point charges

Electric dipole



$$V = \frac{Q}{4\pi\epsilon |\vec{R} - \vec{R}_0|}$$

radius vector of point of observation. $\vec{R}$
 radius vector of source $\vec{R}_0$
 set ref. point ∞

$$Q_1: V_1 = \frac{Q_1}{4\pi\epsilon |\vec{R} - \vec{R}_{01}|}$$

$$= \frac{Q}{4\pi\epsilon \sqrt{x^2 + y^2 + (z - \frac{d}{2})^2}}$$

$$Q_2: V_2 = \frac{-Q_1}{4\pi\epsilon |\vec{R} - \vec{R}_{02}|}$$

$$= \frac{-Q_1}{4\pi\epsilon \sqrt{x^2 + y^2 + (z + \frac{d}{2})^2}}$$

$$V = V_1 + V_2$$

Superposition!

$$Q_1 (0, 0, \frac{d}{2})$$

$$P(x, y, z)$$

$$\vec{R} = \hat{x}x + \hat{y}y + \hat{z}z$$

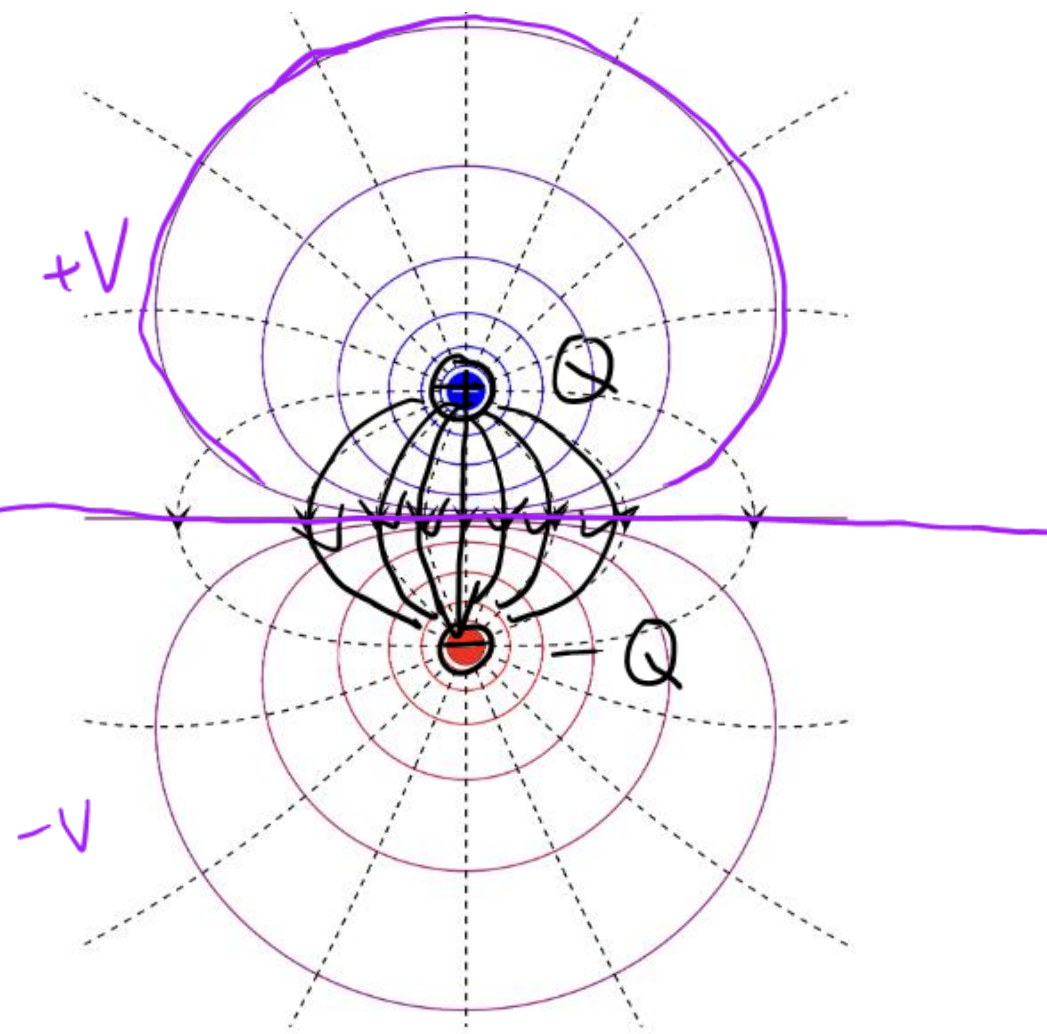
$$\vec{R}_{01} = \hat{z} \frac{d}{2}$$

$$\vec{R} - \vec{R}_{01} = \hat{x}x + \hat{y}y + \hat{z}(z - \frac{d}{2})$$

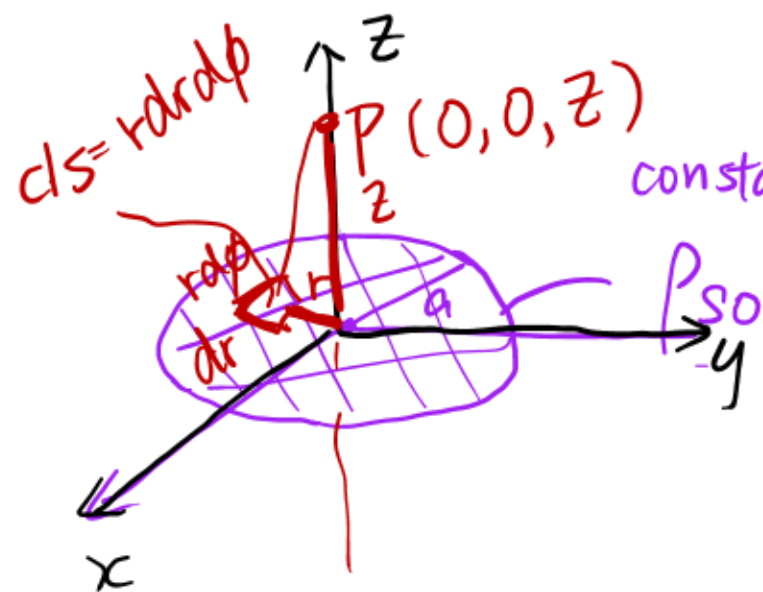
$$Q_2 (0, 0, -\frac{d}{2})$$

$$\vec{R} - \vec{R}_{02} = \hat{x}x + \hat{y}y + \hat{z}(z + \frac{d}{2})$$

$$V_{\infty} = 0$$



Electric potential due to charge distribution



constant surface
charge density

$$0 \leq r \leq a$$

$$0 \leq \phi < 2\pi$$

$$z = 0$$

Point charge Q

$$V = \frac{Q}{4\pi\epsilon |\vec{R} - \vec{R}_0|}$$

given ref. point at ∞
find V on the z -axis.

each ds carry $\rho_{so} ds$, the radius vector $\vec{R}_0 = \hat{r} r$

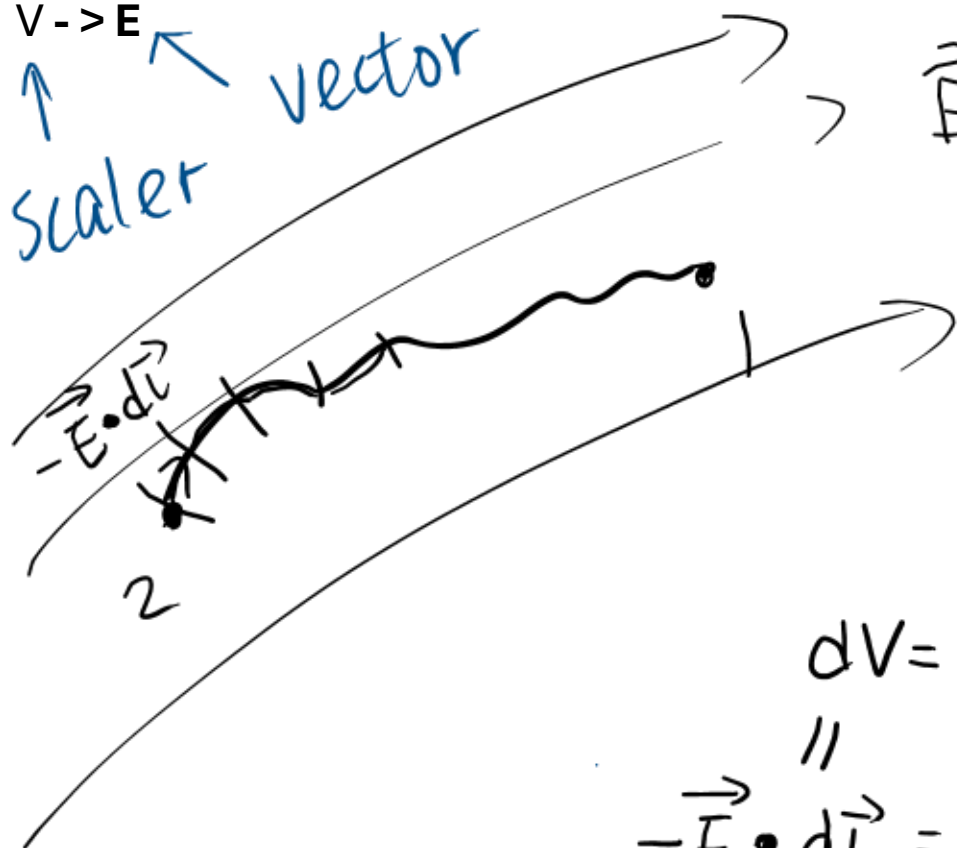
$$dV = \frac{\rho_{so} ds}{4\pi\epsilon |\vec{R} - \vec{R}_0|} = \frac{\rho_{so} ds}{4\pi\epsilon \sqrt{r^2 + z^2}}$$

distance between P and source point.

$$V = \int_{\phi=0}^{2\pi} \int_{r=0}^a \frac{\rho_{so} r dr d\phi}{4\pi\epsilon \sqrt{r^2 + z^2}} = \frac{\rho_{so}}{4\pi\epsilon} \int_{\phi=0}^{2\pi} d\phi \int_{r=0}^a \frac{r dr}{\sqrt{r^2 + z^2}}$$

$$= \frac{\rho_{so}}{2\epsilon} (\sqrt{a^2 + z^2} - |z|)$$

$V \rightarrow E$
 ↑
 scalar vector



$$V_1 - V_2 = - \int_2^1 \vec{E} \cdot d\vec{l}$$

$$dV = - \vec{E} \cdot d\vec{l}$$

$$\vec{E} = \hat{x}E_x + \hat{y}E_y + \hat{z}E_z$$

$$d\vec{l} = \hat{x}dx + \hat{y}dy + \hat{z}dz$$

$$dV = \frac{\partial V}{\partial x} dx + \frac{\partial V}{\partial y} dy + \frac{\partial V}{\partial z} dz$$

$$-\vec{E} \cdot d\vec{l} = -E_x dx - E_y dy - E_z dz$$

$$E_x = - \frac{\partial V}{\partial x}$$

$$E_y = - \frac{\partial V}{\partial y}$$

$$E_z = - \frac{\partial V}{\partial z}$$

$$- \left(\hat{x} \frac{\partial V}{\partial x} + \hat{y} \frac{\partial V}{\partial y} + \hat{z} \frac{\partial V}{\partial z} \right)$$

$$\hat{x}E_x + \hat{y}E_y + \hat{z}E_z$$

$$\vec{E} = - \nabla V$$

$\vec{E}$ points from high V to low V .

In-class problems

Clean Version

4.33 Show that the electric potential difference V_{12} between two points in air at radial distances r_1 and r_2 from an infinite line of charge with density ρ_L along the z axis is $V_{12} = (\rho_L / 2\pi\epsilon_0) \ln(r_2 / r_1)$.

a) You know that you will first need the electric field from an infinite line of charge with density ρ_L . You could just look up the answer in the summary of chapter 4 in the textbook ($\vec{E} = \hat{r} \frac{\rho_L}{2\pi\epsilon r}$), but can you obtain this result quickly using Gauss's law? You should be able to do this with a diagram and 2 to 4 lines of math.

b) Calculate the potential difference between the two points.

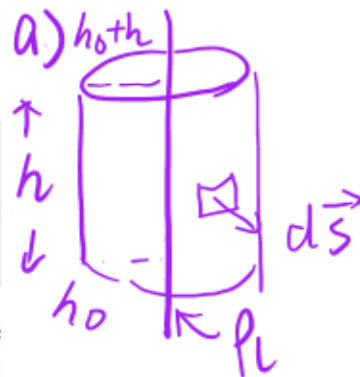
c) What would be the most convenient reference point for this problem? Does it actually matter where you choose your reference?

b.
$$V_{12} = V_1 - V_2 = - \int_2^1 \vec{E} \cdot d\vec{l}$$

$$= - \int_{r_2}^{r_1} \frac{\rho_L}{2\pi\epsilon_0 r} dr = \frac{\rho_L}{2\pi\epsilon} \ln \frac{r_2}{r_1}$$

In cylindrical coordinate system

$$d\vec{l} = \hat{r} dr + \hat{\phi} r d\phi + \hat{z} dz$$



cylindrically symmetric charge distribution

$$\oint \vec{D} \cdot d\vec{S} = Q_{enc}$$

$$\vec{D} = \hat{r} D_r \quad d\vec{S} = \hat{r} r d\phi dz$$

$$\oint \vec{D} \cdot d\vec{S} = \int_{z=h_0}^{h_0+h} \int_{\phi=0}^{2\pi} D_r r d\phi dz$$

$$= D_r \cdot 2\pi r h$$

$$D_r \cdot 2\pi r h = \rho_L h$$

$$D_r = \frac{\rho_L}{2\pi r} \quad E_r = \frac{\rho_L}{2\pi\epsilon_0 r}$$

$$\vec{E} = \hat{r} \frac{\rho_L}{2\pi\epsilon_0 r}$$

In-class problems

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- a) You know that you will first need the electric field from an infinite line of charge with density ρ_ℓ . You could just look up the answer in the summary of chapter 4 in the textbook ($\vec{E} = \hat{r} \frac{\rho_\ell}{2\pi\epsilon r}$), but can you obtain this result quickly using Gauss's law? You should be able to do this with a diagram and 2 to 4 lines of math.
- b) Calculate the potential difference between the two points.
- c) What would be the most convenient reference point for this problem? Does it actually matter where you choose your reference?

$$V - V_{\text{ref}} = \frac{\rho_\ell}{2\pi\epsilon} \ln \frac{r_{\text{ref}}}{r}$$

if $r_{\text{ref}} = \infty$ V is always ∞ NOT OK!

if $r_{\text{ref}} = 0$ V is always $-\infty$ NOT OK!

So where should we put reference point?

Anywhere you want with a finite r_{ref} .

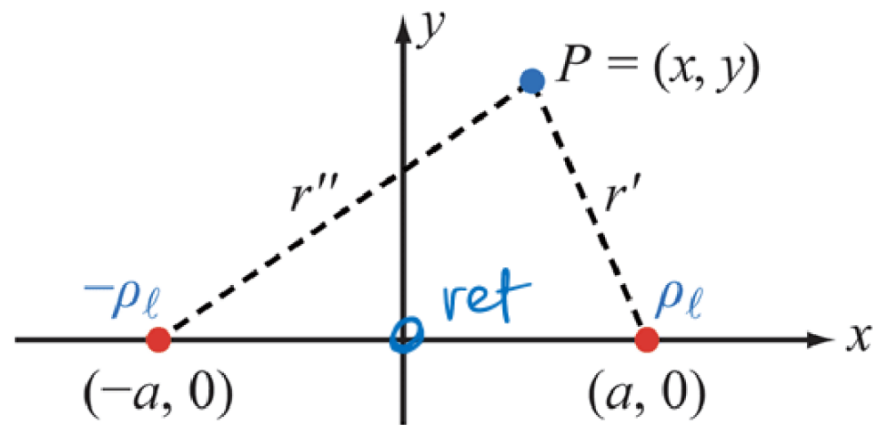
and in your expression, you have to carry r_{ref} info around ☹

C. When we have finite charge distribution, we choose ∞ as a ref. point, it helps clean up the V expression.

BUT! We cannot take ∞ as a ref. point for infinite charge distribution.

As shown in Fig. P4.37, two infinite lines of charge, both parallel to the z axis, lie in the x - z plane: one with density ρ_ℓ and located at $x = a$ and the other with density $-\rho_\ell$ and located at $x = -a$. Obtain an expression for the electric potential $V(x, y)$ at a point $P = (x, y)$ relative to the potential at the origin.

Figure P4.37 Problem 4.37.



"Superposition"

from previous example.

for a single infinite line of charge.

$$V = \frac{\rho_\ell}{2\pi\epsilon} \ln \frac{r_{\text{ref}}}{r}$$

← distance from source to reference point

← distance from source to point of observation.

$P(x, y)$

Right wire $(a, 0)$

Distance from P to wire $\sqrt{(x-a)^2 + y^2}$

Pick origin as the ref. point.

Right wire contributes

$$V_{\text{right}} = \frac{\rho_\ell}{2\pi\epsilon} \ln \frac{a}{\sqrt{(x-a)^2 + y^2}}$$

Left wire contributes

$$V_{\text{left}} = \frac{-\rho_\ell}{2\pi\epsilon} \ln \frac{a}{\sqrt{(x+a)^2 + y^2}}$$

$$V_{\text{total}} = V_{\text{right}} + V_{\text{left}} = \frac{\rho_\ell}{2\pi\epsilon} \ln \frac{\sqrt{(x+a)^2 + y^2}}{\sqrt{(x-a)^2 + y^2}} = \frac{\rho_\ell}{4\pi\epsilon} \ln \frac{(x+a)^2 + y^2}{(x-a)^2 + y^2}$$

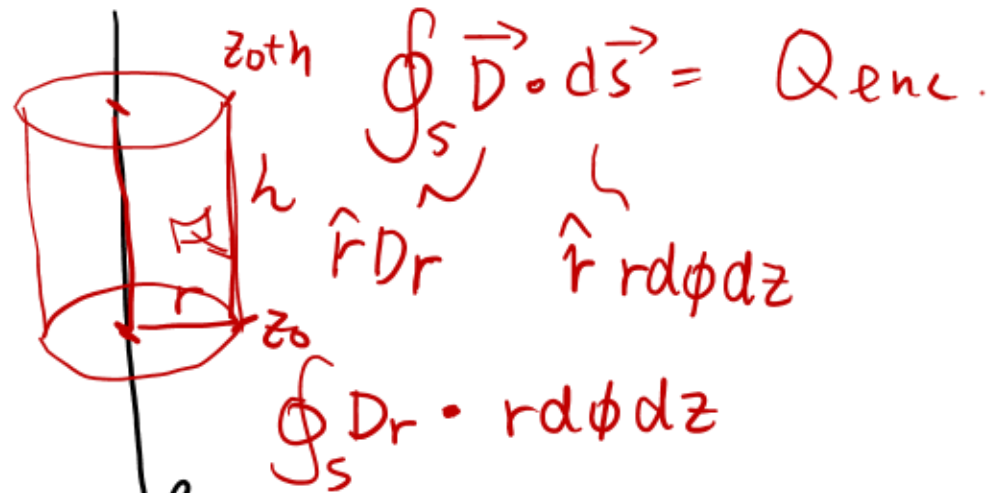
In-class problems

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b) Calculate the potential difference between the two points.

c) What would be the most convenient reference point for this problem? Does it actually matter where you choose your reference?



$$\oint_S \vec{D} \cdot d\vec{s} = Q_{enc.}$$

$$\oint_S \vec{D} \cdot \hat{r} r d\phi dz$$

$$D_r \cdot 2\pi r \cdot h = \rho_L h$$

$$D_r = \frac{\rho_L}{2\pi r}$$

$$E_r = \frac{D_r}{\epsilon_0} = \frac{\rho_L}{2\pi\epsilon_0 r}$$

$$V_{Ac} = V_{Bd}$$

$$\vec{E} = \hat{r} \frac{\rho_L}{2\pi\epsilon_0 r}$$

$$b) V_{12} = V_1 - V_2 = - \int_2^1 \vec{E} \cdot d\vec{l}$$

$$= - \int_{r_2}^{r_1} \frac{\rho_L}{2\pi\epsilon_0 r} dr$$

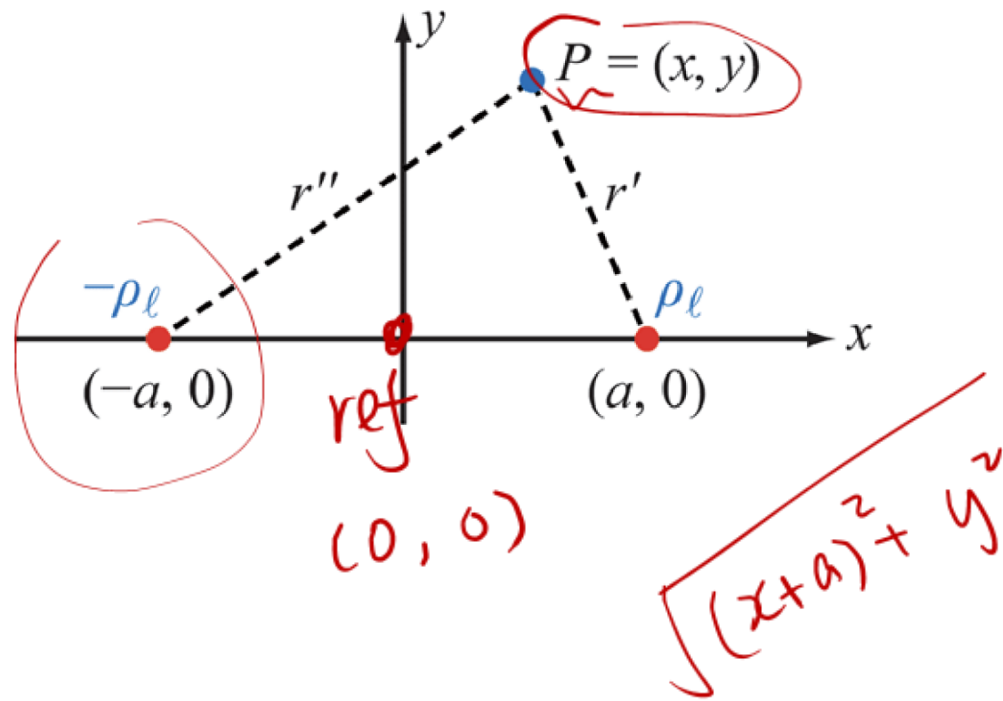
$$= - \frac{\rho_L}{2\pi\epsilon_0} \int_{r_2}^{r_1} \frac{1}{r} dr$$

$$= - \frac{\rho_L}{2\pi\epsilon_0} (\ln r_1 - \ln r_2) = \frac{\rho_L}{2\pi\epsilon_0} (\ln r_2 - \ln r_1)$$

$$= \boxed{\frac{\rho_L}{2\pi\epsilon_0} \ln \frac{r_2}{r_1}}$$

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Figure P4.37 Problem 4.37.



$$V = \frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{r_{\text{ref}}}{r}$$

left wire $V_{\text{left}} = \frac{-\rho_\ell}{2\pi\epsilon_0} \ln \frac{a}{\sqrt{(x+a)^2 + y^2}}$

right wire $V_{\text{right}} = \frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{a}{\sqrt{(x-a)^2 + y^2}}$

$$V = V_{\text{left}} + V_{\text{right}} = \frac{\rho_\ell}{2\pi\epsilon_0} \left[\ln \frac{a}{\sqrt{(x-a)^2 + y^2}} - \ln \frac{a}{\sqrt{(x+a)^2 + y^2}} \right]$$

$$= \frac{\rho_\ell}{2\pi\epsilon_0} \ln \frac{\sqrt{(x+a)^2 + y^2}}{\sqrt{(x-a)^2 + y^2}}$$